Fuel Stop — Sprint 1 PRD, Revision 5: Launch

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Revision 5, 19 October 2026. Supersedes Revision 4 (19 October 2026).

> **Fuel Stop launched on 19 October 2026** at **https://map-website-snowy.vercel.app**.
> Revisions 1 (`SilvaHefferon_ProjectBrief.pdf`), 2 and 3 (`Map Website_Sprint 1_PRD 2.md`, `… 3.md`) and 4 (`Fuel Stop_Sprint 1_PRD 4.md`) are unchanged on disk as the record of earlier plans. Where they disagree with this document, this one is current.

What changed since Revision 4

Area	Revision 4	Now
Status	Live, still building	**Launched**
Restroom-friendly routes	Must-have, not built	**Deprioritized for time;** no longer a must-have
Local specialties	—	**Built** (PRs #4 and #5): restaurants known for a Boston signature dish are labelled, and count as "Local" cuisine
Brand	Colour logo	Plus an **all-white** version; the browser-tab icon uses it

Every remaining must-have is built.

The idea and the user

Fuel Stop is an interactive web map for "foodie runners", people new to a city who care about both their run and their next meal. It brings running routes, public restrooms and authentic local restaurants together on one map, and runs from one to the other.

Scope for this sprint

Must-haves: all built

#	Feature	State
1	Route generator: adjustable distance, start point, loop or run to a point	**Built**
2	Elevation as an input: flattest / balanced / hilliest	**Built**
3	Restaurant finder, filtered by price and "recognised" picks	**Built**, prices in dollars

#	Feature	State
4	Restaurant quality from established sources, not star ratings alone	**Built**, plus local specialties
5	Public restroom finder	**Built** (117 restrooms)
6	Run to a restaurant	**Built**

Deprioritized: restroom-friendly routes

Restroom-friendly routes were a must-have in Revisions 3 and 4. **We deprioritized them for time** to launch this sprint. Restrooms are still on the map (feature 5); routes just don't choose themselves by them yet.

The design stays on record in Revisions 3 and 4, for whoever picks it up: - A restroom serves the route within about 150 m. - Rank routes by the longest stretch without one, then by how many they pass. - Never promise a restroom is open. OpenStreetMap has opening hours for only 8 of the 117.

The code now has a natural place for it: hilliness already ranks candidate routes near the requested distance, and restrooms would rank them first.

Out of scope

Recommending restaurants *along* a route, live navigation or GPS tracking, user accounts and saved routes, user-submitted reviews, and a mobile app (the page works on a phone-sized screen).

What launched

Routes

- **Loop or run to a point.** *Loop* starts and ends at the start point. *To a point* runs one way to a destination picked on the map, or to a restaurant via **Run here** in the Restaurants tab. Out-and-back left the page in Revision 4; the server still supports it.
- **Distance within 500 m** at every length, not a percentage, so a 20 km run can't drift 2 km. The results panel shows the difference in metres, and says so when no route lands within 500 m.
- **Flattest / balanced / hilliest** choose among the candidates within 500 m by their climb on a USGS elevation grid. Distance comes first. The page says when routes here barely differ.
- **Runs to a point loop round** through an arc of waypoints, not out and back along a hairpin. Within 500 m of the direct route, the direct route is the run. More than 500 m short of it, the page warns that the point is too far for that distance.
- **Results:** distance, estimated time, arrival time for runs to a point, and climb.
- **Map clicks** move the start only on the Routes tab, set the destination while picking, and never act on restaurant pins.

Restaurants

- **Ranked recommendations** from recognised local sources. Scoring is below.
- **Prices in dollars**, e.g. `$$$ ($50–75)`, from the same source as the level: Google's range, Yelp's band or Michelin's band (PR #3).

- **Local specialties** (PR #4). A restaurant gets a "Local specialty: Lobster roll" tag when what's written about it says it is *known for* one of Boston's 14 signature dishes, not merely that it serves one.
- 19 restaurants are labelled, e.g. Neptune Oyster, Saltie Girl and Union Oyster House.
- Each label keeps a word-for-word quote from its source. The tag shows the quote on hover and links to the source.
- Labels were made offline with Claude and checked by hand. The site itself never calls Claude.
- A specialty adds 3 points but never makes a place "recognised" on its own, because a famous lobster spot can still be a tourist trap.
- The **"Local" cuisine filter** includes these places too (PR #5).

Restrooms

117 public restrooms in Boston, each labelled with its name or its nearest intersection. Airport restrooms are excluded.

Brand

Colour	Hex	Used for
Green	`#1b4436`	Header text, tabs, buttons, the route line
Orange	`#e8743f`	Recognised restaurants, run destinations
Cream	`#f8f6f1`	Page and panel backgrounds

The logo, an orange runner's dot and the green path that becomes a fork, is in `shared/brand/` as transparent SVG: - colour, with and without the name: `logo.svg`, `mark.svg` - all white: `logo-white.svg`, `mark-white.svg`

The header uses the colour mark; only the browser-tab icon uses the white one.

Restaurant quality: the scoring model

Weights live in `restaurants/scoring.py`, and `restaurants/README.md` has the full model.

Ingredient	Points
Michelin star / Bib Gourmand / Guide	10 / 8 / 6
Eater 38 / Eater Heatmap	8 / 4
City or tourism-board food guide	4 each, max 2
TheFork, Zomato	2
Local food creator list	2 each, max 3
Each extra independent publisher agreeing	+2
Local specialty *(new)*	**3**, once per restaurant
Google rating, review-adjusted, above 4.4	up to 3

Ingredient	Points
Yelp rating, review-adjusted, above 4.0	up to 3

Ratings can't outrank critics: Google and Yelp together cap at 6, below a single Michelin star.

How it is built

- **One map, owned by `shared/`.** Features add layers through `view.addLayer`. They talk to each other only through `view.emit` / `view.on`. The one message so far is `restaurants:run-to`, sent by **Run here**.
- **One small Python server**, standard library only. It runs locally as `python server/app.py` and on Vercel through `api/index.py`. Static files are served directly. API keys live in `.env` locally and in the Vercel project's settings, never in the repo.
- **Vercel** deploys from `main`. The city's restaurant data ships with the site (`data/cache-seed/`), because fetching it fresh takes longer than Vercel allows a request.
- **Tests:** 192, all passing at launch.

Cost and dependencies

- **No card is needed.** Restaurants use OpenStreetMap; routing uses OpenRouteService's free tier; Vercel's free plan hosts the site. Local specialties were labelled without the paid Claude API.
- **The routing quota is the constraint.** OpenRouteService's free plan allows 2,000 requests a day, shared by every visitor, at up to 6 per route. It ran out on launch day. **HeiGIT's free academic plan (10,000 a day, with an MIT email) is the fix.**

Success criteria

Measured after launch:

1. **Restaurants.** A local expert approves **at least 90%** of the recommendation list.
2. **Routes.** A test runner runs a generated route and confirms it works as planned.
3. **Run to a restaurant.** A test runner reaches a chosen restaurant on a generated route within **500 m** of the requested distance, with no impassable sections.

Restroom-friendly routes' criterion is dropped with the feature.

Still open: "90% of the list" has no list length; the expert and the test runner are still to be recruited.

Known issues at launch

Issue	Impact	Fix
Routing quota spent on 19 October	No routes until it resets; the page says so	HeiGIT academic plan; find what spent it
Rate limits are counted per Vercel instance	The shared quota is guarded less tightly than on one server	A shared counter, or the academic plan's headroom

Issue	Impact	Fix
Vercel twice missed or stalled deploys on 19 October	Pushes reached the site only after a manual deploy	Check Settings → Git; watch the next pushes
White tab icon on light-theme browsers	Hard to see	A theme-aware icon, if wanted
500 m tolerance on short runs	400 m out on a 2 km run passes	"500 m or 10%, whichever is smaller"
Long runs to nearby places	Can land outside 500 m; the page warns	More attempts, at more quota
Elevation over bridges (Mass Pike, I-93)	Reads high	Follow the deck where OpenStreetMap marks a bridge

Open questions

#	Question	Status
1	A permanent name	Closed — Fuel Stop
2	Recruit the local food expert	Open
3	Recruit the test runner	Open
5	Recognition data: sources to verify	Open — each entry copied from its published source with a link
7	How many results does the expert review?	Open
10	When distance and hilliness conflict, which wins?	Closed — distance (within 500 m), then hills
11, 12	Restroom distance and hours	Deferred with restroom-friendly routes
13	Acceptable detour shapes	Closed — loops, not hairpins
14	What a map click does	Closed
15	Pedro's agreement to the cross-feature hook and colour changes	Open
16	Routing quota	Open — academic plan
17	A public link	Closed — launched 19 October
18	Tolerance on short runs	Open
19	Vercel's per-instance rate limits	Open

Status at launch

	State
Live site	https://map-website-snowy.vercel.app, running `main`
`main`	PRs #1–#5 merged. 192 tests passing
Routing	Down until OpenRouteService's daily quota resets

After launch:

1. Apply for HeiGIT's academic plan, and check what spent the quota.
2. Confirm with Pedro: the cross-feature hook and the colour changes in his files.
3. Recruit the food expert and the test runner, set the review list's length, and measure the success criteria.
4. Restroom-friendly routes, when there's time.